Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/17/2025
PI Name:	Settles, Andrew Mark Ph.D.		
Project Title:	Feasibility of Synthetic Biology Countermeasures for Human Exploration Beyond Low Earth Orbit		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-LEIA E.10. Lunar Explorer Instrument for Space Biology Applications
Start Date:	12/01/2021	**End Date:**	09/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2024 per F. Hernandez/ARC (Ed., 12/12/23) NOTE: Project dates (POP) changed; now 12/1/2021-11/30/2023 per F. Hernandez/ARC (Ed., 1/19/22)		
Key Personnel Changes/Previous PI:	Dr. Aditya Hindupur resigned from his Science co-I position in April 2022 to take an industry position.		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial production of bioactive compounds, such as vitamins or pharmaceuticals, can reduce risks for deep-space crewed missions. Yeast are excellent chassis organisms to express countermeasure products due to their long shelf-life viability. Yeast have robust synthetic biology technology to transfer whole biosynthesis pathways for synthesis of desired products. NASA has invested in yeast production of micronutrients that are known to have a short shelf-life in prepackaged foods, with demonstrated success on the International Space Station. However, low-Earth orbit does not test yeast for resistance to higher radiation levels and the more extreme environment of deep space. The goal of this proposal is to develop methods to preserve, grow, and measure production of desired synthetic biology products from edible yeast, using the BioSensor platform. BioSensor automates yeast culture activation and monitors growth with light absorbance of specific wavelengths produced by light emitting diodes (LED). The project goal is to expand the capability of BioSensor to enable monitoring synthetic biology traits along with cell growth and metabolism. The central objectives are: 1) Develop methods to predict synthetic biology production traits, namely carotenoids and recombinant proteins, using multivariate statistical models based on three wavelength light absorbance. We anticipate that the BioSensor platform will need to be modified to replace one of the current wavelengths to a blue LED; 2) Yeast may be overly sensitive to deep space radiation, and we will engineer carotenoid producing strains to express a DNA damage protection protein from tardigrades; 3) Non-conventional, yeast species may be more efficient for recombinant protein expression in deep space conditions. We will engineer three edible species to produce a target protein that absorbs blue light, to enable monitoring of recombinant protein and carotenoids in the same BioSensor device; and 4) Determine strain and media storage conditions as well as test the multiwavelength light monitoring strategy to establish the requirements and methodology for a future lunar surface mission. The project is expected to advance the remote sensing technology in the BioSensor platform. A future flight experiment is expected to develop fundamental knowledge on the effects of deep space on protein expression and metabolite production. In addition, the project directly addresses critical gaps to advance crewed missions in deep space exploration. The specific carotenoids, beta-carotene and zeaxanthin, are important micronutrients to prevent macular degeneration and have been identified as potential countermeasures to maintain vision in astronauts during deep space missions. Recombinant protein expression in non-conventional yeast will demonstrate feasibility for production of bioactive protein therapies in deep space missions. These synthetic biology approaches are critical to provide products that are sensitive to radiation and have short shelf lives in prepackaged foods and medicines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microbial production of bioactive compounds, such as vitamins or pharmaceuticals, can reduce risks for deep-space crewed missions. Edible yeasts are chassis organisms that are highly amenable to synthetic biology and can produce a vast array of useful compounds including vitamins and pharmaceuticals. This project is developing yeast genetic strains and remote sensing methods to study the impact of lunar surface radiation on the growth and productivity of engineered yeast. The project will be testing methods to mitigate the impact of oxidative stress in the cell on productivity of desired compounds. The fundamental knowledge developed in the project will contribute to a better understanding of cellular reactions to unfavorable environments and may reveal new approaches to improve microbial biomanufacturing when cells are experiencing environmental stress.
Task Progress:	This project developed engineered yeast to be assayed for performance in extreme deep-space environments using the BioSensor microfluidics system that was developed for the BioSentinel CubeSat as part of Artemis I. The BioSensor uses light emitting diodes (LED) at three wavelengths to allow measurements of cell growth and color changes in yeast cultures. The project research activities complement and directly support a NASA "Payloads and Research Investigations on the Surface of the Moon" project entitled, “Lunar Explorer Instrument for space biology Applications (LEIA).” The LEIA suite of instruments includes the BioSensor, a charged particle, linear energy transfer spectrometer, and fast neutron detector, which will be delivered to the lunar surface as part of a Commercial Lunar Payload Services (CLPS) mission. The two radiation detectors will allow direct correlation between radiation exposure with yeast growth and production in the BioSensor. This project made significance progress on all four specific aims as follows. Aim 1) Develop methods to predict synthetic biology production traits, namely carotenoids and recombinant proteins, using multivariate statistical models based on three wavelength light absorbance. We determined that optical density measurements at 465 nm and 850 nm can be used to detect beta-carotene production in microtiter plate formats. Mixing experiments with non-producing and carotenoid producing yeast strains showed high accuracy calibrations of 465 nm and 850 nm absorbance to dry cell weight equivalents of beta-carotene production levels. We also engineered baker’s yeast to express Enhanced Green-Cyan Fluorescent Protein (ECGFP) and determined that this protein does not absorb sufficient 465 nm light to gauge engineered protein production. These experiments provided evidence that the spectroscopy approaches planned for carotenoid detection in lunar surface BioSensor experiments as part of the LEIA mission are feasible. Aim 2) Engineer carotenoid-expressing yeast strains that enhance sensitivity or resistance to expected lunar surface environment stressors. We generated twelve yeast strains that are expected to alter sensitivity to abiotic stresses such as reactive oxygen species (ROS) or radiation. Four strains were confirmed to be more sensitive to long-term storage of dry cells in ambient, laboratory conditions. Five of the yeast strains tested expression of Dsup or CAHS genes from tardigrades, which have been shown to enhance radiation or desiccation tolerance in plants, animals, or yeast. 3) Engineer a non-conventional yeast species to express a blue light compatible marker for synthetic biology-enabled production. We developed three single-locus beta-carotene biosynthesis pathway constructs that were integrated into multiple yeast species, including three non-conventional yeast species and baker’s yeast. Although the non-conventional yeast species produced carotenoids, desiccation tolerance testing showed that these species were not suitable for the LEIA lunar surface mission. As an alternate approach, we engineered gene disruptions of the rgt2 and snf3 loci in carotenoid-producing baker’s yeast to generate a strain that emulates Crabtree-negative yeast metabolism found in non-conventional yeast species. Preliminary testing of this strain showed higher levels of carotenoid production.

	4) Use BioSensor microfluidics cards to test inoculum, desiccation, and spectroscopy methods to measure cell growth and product formation. Initial tests with BioSensor microfluidics cards in ground support equipment (GSE) showed feasibility to detect beta-carotene production using visible light absorbance at 465 nm and NIR absorbance at 850 nm. However, our primary beta-carotene producing strain was not biocompatible with the media planned for the LEIA lunar surface mission. We engineered additional baker's yeast strains in the BioSentinel genetic background with BioSensor flight heritage. These new carotenoid producing strains were found to be biocompatible with the LEIA flight media and produced sufficient carotenoids to be measurable with the optical assay planned for the lunar surface mission.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
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Abstracts for Journals and Proceedings	Settles AM. "Synthetic biology to support human exploration of deep space." 2024 World Congress on In Vitro Biology, St. Louis, Missouri, June 8-12, 2024. Abstracts. 2024 World Congress on In Vitro Biology, St. Louis, Missouri, June 8-12, 2024. , Jun-2024
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